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ESIR

Focus paper: A European Model for Artificial Intelligence



Independent Expert Report

Research and Innovation

A European Model for Artificial Intelligence

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EUROPEAN COMMISSION

A European Model for Artificial intelligence

A Focus paper by members of the Expert group on the Economic and Societal Impact of Research and Innovation

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TOWARDS A EUROPEAN MODEL FOR ARTIFICIAL INTELLIGENCE

The European Union faces the dual challenge of harnessing Artificial Intelligence (AI)[1] to stay globally competitive while championing ethical considerations, data privacy, transparency, and accountability. This focus paper digs into the key aspects of creating a unique European approach to AI, tackling the challenges it faces, and leveraging its strengths to overcome them.

The paper discusses the current landscape of AI in Europe. It is no surprise that Europe faces some tough structural challenges, like building robust infrastructures, developing advanced skills, retaining talent, and creating comprehensive data strategies. This comes in the context of external challenges such as global crises and the need for Open Strategic Autonomy, which are shaping the geopolitical and economic landscapes of AI advancement.

The paper also explores initiatives that have laid the groundwork for a more cohesive and powerful AI strategy in Europe. ESIR experts examine areas like EU leadership in science, the role of the public sector in building self-sovereign infrastructures, and how to invest wisely while ensuring compliance with a human-centered legal framework. Europe must build a competitive AI ecosystem that also drives our societal and environmental goals. This might involve ambitious research and innovation policies, like an AI moonshot, or focusing on smart specializations that align with other EU policies and objectives.

1. Context: current European initiatives and structural and external challenges

1.1. Structural challenges: infrastructures, skills and data

Europe is lagging behind in the global AI landscape. Despite having a massive AI talent pool and strong fundamental research capacity, Europe has struggled to keep pace with the United States and China in terms of product development. Historical examples, such as the Hungarian social network iWiW and European e-commerce platforms, highlight both Europe's potential and past missed opportunities. The lack of a unified digital single market has hindered Europe's ability to lead in internet applications and now in AI, with far-reaching consequences. This failure to lead in AI not only has significant economic implications but also limits Europe's ability to ensure economic security, reduce dependencies on its main competitors, and direct AI development towards broader economic, societal, and environmental goals. To change this trajectory, Europe must prioritize building a cohesive AI strategy that leverages its strengths and addresses its weaknesses. The good news is that this is possible but will require a clear vision and unprecedented investments.

The dominance of US and Chinese companies in AI is evident in the data, as illustrated by Figure 1, which ranks the world's largest publicly listed companies in AI-relevant

[1] "Artificial Intelligence refers to systems that display intelligent behaviour by analysing their environment and taking action — with some degree of autonomy — to achieve specific goals." Communication from the Commission to the European Parliament, the European Council, the Council, the European Economic And Social Committee and the Committee of the Regions - Coordinated Plan on Artificial Intelligence – 2018 - https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=COM:2018:795:FIN

technology sectors by market capitalization. Companies like Alphabet, Apple, Amazon, Meta, Microsoft, Nvidia, and Tesla - the so-called "magnificent seven" - have excelled in creating AI-powered recommender systems, generating massive value and innovation. However, this dominance also raises concerns about the potential for echo chambers, exacerbating inequalities and biases, and undermining societal cohesion and democratic processes.

Europe's absence in this domain means that **citizens are forced to accept cultural and ethical standards embedded in AI developed elsewhere**. This underscores the urgent need for European-focused AI platforms and policies that reflect European values and priorities.

The remarkable growth of companies like Nvidia, driven by their ability to produce crucial AI-enabling chips, has contributed significantly to the **divergence between the EU and the US in terms of productivity and growth** over the past decade. At the time of writing this paper, Nvidia just passed Apple as the world's second-most valuable company, now sitting behind only Microsoft. These companies have successfully captured the value generated by the remainder of the economy, leveraging personal data and increasingly, industrial value. The result is stunning: the EU27's GDP has plummeted from 91% to 65% of the US GDP between 2013 and 2023, according to the Financial Times (2023). This trend has significant implications for Europe's economic security and competitiveness.

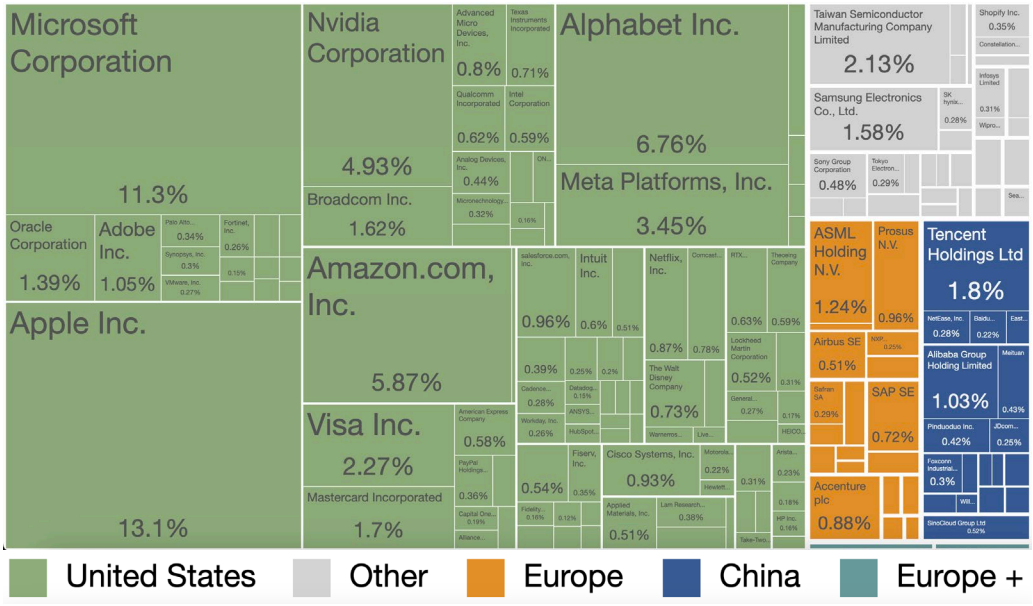


Figure 1 - World's largest tech companies by market capitalization as of July 2023 (Source: author's own work)

The rise of big tech in the United States is also the result of these companies' ability to leverage the data economy by capturing the attention of the end users, which ultimately resulted in a high degree of concentration in key domains of cyberspace. **Private control of large data sets by a few companies, however, while creating opportunities for application developers also in Europe, might constrain the development of new AI-powered applications, especially in Europe, where both the private and public sectors could greatly benefit from AI**; and lead to exclusive reliance on often "black box" AI models

developed outside of the EU, of which European companies are becoming mere deployers, with little or no access to the underlying technology.

Turning the tide is not going to be easy, yet Europe is not starting from zero. On the one hand, the uptake of AI among businesses and public administrations has been rather poor. **Even in the domain of regulation, traditionally Europe's pride, things are challenging due to the fast pace of AI development, which in turn makes it difficult to balance regulation with fostering innovation**. This is true in particular when it comes to creating the preconditions for adequate data governance and stewardship and providing legal certainty to start-ups willing to develop AI-powered solutions. The notion of "regulatory sandboxes" in AI to test innovative products or services that challenge existing legal frameworks enabling participating firms to obtain a waiver from specific legal provisions or compliance processes to innovate (European Parliament, OECD, 2023) resembles today more regulatory "quicksandboxes". The regulation of AI is hence continuously subject to debate, revision and further development, such as with the recent AI Act[2]. The extent to which calls for further regulation are part of incumbents' own strategies to increase the value of their data stock or to prevent newcomers from entering their markets is very much open for debate (Burgelman, 2023).

Yet on the other hand, **Europe is the land of cutting-edge supercomputers, as well as a major source of talent**, forming more AI-specialised researchers than the United States every year. To avoid that these resources are under-utilised (in the case of supercomputers) or end up serving other markets (talent, often attracted by top U.S. destinations), *ad hoc* policies will be needed. Plus, these policies will need to be complemented by massive efforts to improve data availability, security and governance. Among potential flexible framework needs, an adapted legal framework on data including a legislative framework for European data governance and horizontal rules increasing legal certainty on data access and use would be useful. As Barend Mons (2023) highlighted "good practices in data stewardship and analysis could be more effective than regulation", hence making a plea to make such data "Fully AI Ready (an alternative wordplay on the original meaning of the FAIR acronym; **F**indable, **A**ccessible, **I**nteroperable and **R**eusable for machines as well as people)", with the primary purpose to further research in AI-applications. **European authorities should prioritise making the data used for AI training transparent or public,** instead of focusing solely on regulation. This lack of access hinders Europe's ability to catch up in AI applications and future developments. In contrast, US and Chinese tech giants have amassed vast data sets, propelling their AI research and development, while also raising concerns about the uncontrolled pace of AI advancements. This transparency could foster broader applications and improvements in AI, but it faces resistance from big tech firms who view their data collection as a critical investment.

In crafting a future industrial policy on AI for Europe, regional specialisations and complementarities should be considered. **AI innovation and talent are very concentrated in only a handful of European cities.** When looking at patents over the past 8 years (Figure 2), it is clear that four European regions account for more than 25% of all AI patent applications in Europe. These are Inner London, Ile de France (region of Paris), Noord-Brabant (region of Eindhoven) and Oberbayern (region of Munich). Furthermore, in relative terms (as measured by the revealed comparative advantage) the Bucharest region, Estonia, Budapest region & Brussels show significantly above-average shares in AI patents compared to other European regions. **The takeaway message here is that there is a significant concentration of AI talent and innovation potential in some European cities and regions.** There exists a pressing need for substantial investments in AI research and development, with a keen focus

[2] https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai

on nurturing these hubs and encouraging innovation to foster an environment where European entities can thrive and compete globally.

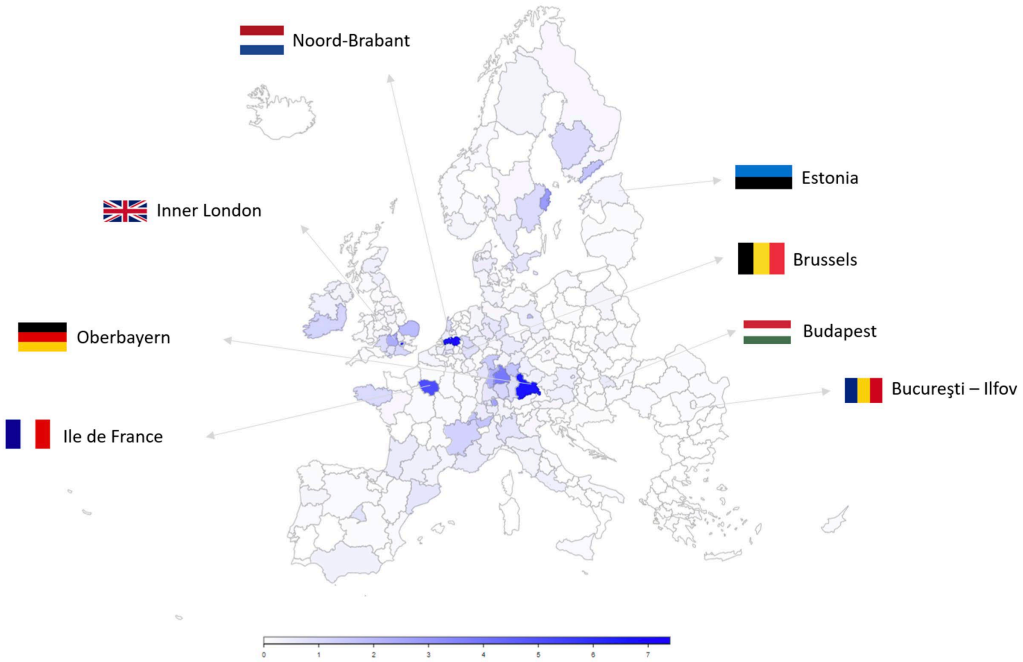


Figure 2 - Share of AI patents in European regions (Source: authors' own work based on the OECD RegPat database, version 2024)

Importantly, existing European hubs are not strongly connected to each other, whether through patents or other collaborations. Analysing inter-regional AI patents between European regions, the vast majority of connections remain within the same country (see Figure 3). But EU AI hubs are rarely within the same countries. Ile de France, for instance, is more connected to Rhone-Alpes than to London. Upper Bavaria is more connected to Mittelfranken than to any other major AI hubs. Noord-Brabant tends to be more connected with other AI hubs. Strong connections exist between Brussels and Zurich, Trier and Luxembourg, and the Prague region and Stockholm. This fragmentation is also visible when it comes to AI policy and initiatives – they tend to be led by Member States and often lack a European component.

As a result, Europe needs much more inter-hub connectivity to produce the complex AI technologies the world needs[3].

[3] For more details see SRIP 2022, Innovation policy in a complex world https://research-and-innovation.ec.europa.eu/system/files/2022-07/ec_rtd_srip-2022-report-chapter-14.pdf

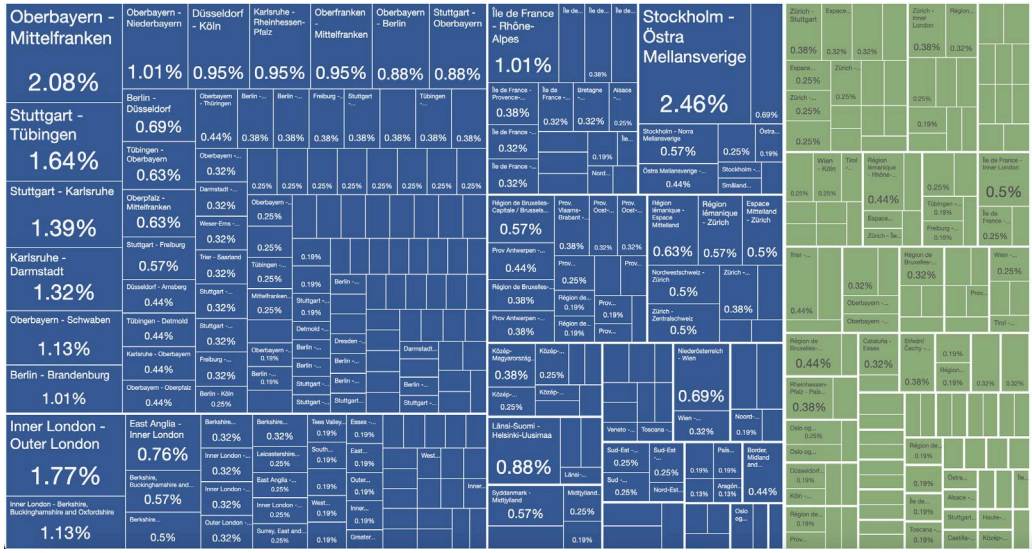


Figure 3 - Share of inter-regional AI collaboration in Europe (patents) (Source: authors' own work based on the OECD RegPat database, version 2024)

Overall, the EU has strong arguments to dispute the global race to develop and harness the potential of AI, but it suffers from structural issues as summarised in Table 1. These must be tackled head on if Europe wants to have a say in the AI domain.

Strengths	Weaknesses
Vibrant start-up ecosystems (Paris, Munich, Berlin) with foundational startups	Access to Venture Capital, especially when compared to competitors in terms of volume and accessibility
Advanced High-Performance Computing infrastructure and data spaces (EuroHPC)	Lack of relation between AI hubs, lack of large-scale data solutions and support
Forward-looking pioneer regulation (AI Act)	“Catching-up mode”, full reliance on foreign AI black box models and data storage
Excellent research ecosystems, with multiple AI research initiatives across the Member States	Uncoordinated approach divided across Member States limiting effectiveness and scale-up

Table 1 - Strengths and weaknesses of the EU in the development of AI

1.2. External challenges: poly-crises and the Open Strategic Autonomy imperative

While the current phase of AI developments is far from over and presents the European Union with major challenges, since November 2022 a new phase of the AI revolution has come of age. It really started with the rise of Large Language Models (LLMs) such as OpenAI's ChatGPT, which reached hundreds of millions of users in about 2 months, becoming the fastest-growing application in the history of the Web. ChatGPT is just one LLM, but other models such as Gemini Ultra, LLaMA3, or Claude demonstrated outstanding capabilities. And in China, homologous models such as the recent Alibaba's Qwen2, quickly appeared on the market, giving rise to what Tencent's CEO defined as a "war of a hundred models".

The economic and social impact of LLMs might be even larger than the previous phase of AI innovation. Recommender systems, for instance, might heavily affect what of the internet world users are allowed to see, but they do not seem to massively automate human tasks. LLMs, on the contrary, have the potential to do exactly that, in a growing array of tasks and activities that were previously thought to be inaccessible to machines. And even more importantly, the two phases of the AI revolution are intimately linked: in other words, those that successfully mastered the previous one, building a customer base, and massive data and computing infrastructure, have an advantage also in embracing the next phase. Not surprisingly, ChatGPT, Claude, or Gemini with their billions of parameters, do not come from France, Hungary or Sweden. **If such a situation continues, the EU might end up in a situation where LLMs trained with European data are designed, developed and deployed by U.S.-based companies and end up replacing jobs in Europe.**

Not being part of the frontrunners in AI development has obvious economic impacts, and also prevents Europe from shaping the development of AI towards desirable societal goals. Recommender systems are very powerful engines to navigate a complex world. Given the vast amount of content available online, it has become essential to have systems in place for searching, filtering, and aligning content with user preferences.AI recommender systems function by analysing a user's past behaviour, preferences, and interactions to suggest content, products, or services. In simple terms, what social media recommender systems do is essentially to personalize content and decide what users see. This is usually based on a technology called collaborative filtering, whose systems filter information sent to the users. This act of filtering information is by nature not neutral, and the design and implementation of these systems can create inequality through the mechanism of preferential attachment, but also polarizing echo chambers. Echo chambers form when users are disproportionally exposed to information and opinions like their own - thereby reinforcing their existing beliefs and shielding them from diverse perspectives. By not having viable European AI platforms, European users must accept cultural and ethical embedding from AI developed in other parts of the world, and this is despite extensive regulatory work. The EU has institutions that are comparatively well aligned with citizens' interests, but this does not mean that ethical AI will automatically emerge from these protective regulations. Such an ethical AI would be the outcome of a complex process that will require new carefully designed R&I policies and instruments.

Simply forcing big-tech giant platform companies to make available the underlying client data used to train the algorithms they developed is likely to be unacceptable to those firms. These relatively young, today incumbent platform companies will claim that they invested heavily in the build-up of their customers' data; and that they were "innovators" in linking up at first sight unrelated data so as to train newly developed algorithms. In short, that their data collection efforts were an essential investment feature accompanying the

development of their digital platforms. For smaller, more specialized AI firms, the issue might be rather different: having access to supercomputing facilities could well be the greatest bottleneck as also highlighted by European Commission president Von der Leyen in her 2023 State of the Union speech.

From a societal perspective, it is clear that the private possession of these huge data sets in the hands of a few big tech giant platform companies will limit the number of possible new AI applications. Particularly in Europe where, as Michael Spence (2023) has argued, productivity gains in many public sectors have been relatively limited over the past decade, it is essential with ageing and resulting labour shortages to fully exploit more AI opportunities in order to keep those services accessible to the public. Sectors affected in particular are health, education, public administration, and human protection and policing.

At the same time, and as illustrated in the case of the Dutch tax authorities, public authorities have a particularly poor track record in training algorithms for more publicly oriented AI applications. Meanwhile, the track record of the big tech platform companies in protecting the privacy of the personal data they collect has improved following the implementation of the EU's data protection regulation, but is far from gaining absolute trust from the public.

Furthermore, the ubiquity and rapid growth in AI models size and deployment implies the environmental impact of this technology will only continue to increase, and diversify, stemming from its use in material for the hardware and energy for the development and training of the AI models[4]. This rebound effect adds a further challenge in the development of an AI based on sustainable principles[5].

Furthermore, a critical challenge for the EU will be the access and control of data of its own citizens, with an estimated 90% of global data centres managed by entities outside its jurisdiction, predominantly in the United States[6]. This situation presents a significant hurdle in the path to leveraging AI effectively and autonomously.

Beyond the issue of global competition, the EU is of course also confronted with global challenges, such as global warming and environmental degradation and the rise of large-scale geopolitical tensions. In this framework, AI has the potential to improve our ability to measure and identify weaknesses, priorities and areas of improvement related to the SDGs, while accelerating their achievement[7].

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[4] Berthelot, Adrien, et al., Estimating the environmental impact of Generative-AI services using an LCA-based methodology, Procedia CIRP, Volume 122, 2024, Pages 707-712, ISSN 2212-8271, https://doi.org/10.1016/j.procir.2024.01.098.

[5] Wu, Carole-Jean, et al. "Sustainable AI: Environmental implications, challenges and opportunities." *Proceedings of Machine Learning and Systems* 4 (2022): 795-813. https://proceedings.mlsys.org/paper_files/paper/2022/file/462211f67c7d858f663355eff93b745e-Paper.pdf

[6] https://www.statista.com/statistics/1228433/data-centers-worldwide-by-country/

[7] SRIP 2022, AI for social good: the way forward, https://research-and-innovation.ec.europa.eu/system/files/2022-07/ec_rtd_srip-2022-report-chapter-11.pdf

1.3. European initiatives

Europe's strategy for AI, nested in the 2020 White Paper, emphasises building an ecosystem of trust, as well as an ecosystem of excellence. This approach aims to create a balanced AI environment that fosters innovation while ensuring the safety and protection of fundamental rights.

This is complemented by a regulatory framework for AI structured on a risk-based approach, differentiating between unacceptable, high, and minimal risks. This framework is designed to provide clear guidance for AI developers and users, while also aiming to reduce administrative and financial burdens on businesses. It divides risks according to a pyramid model (Unacceptable, high-risk, limited risk and minimal/no risk), but the assessment of risks of AI systems poses hurdles on its own, such as the lack of assessment and enforcement mechanisms or the burden of expertise it poses on national regulation agencies for a technology that has capacities for spillovers.

The 2018 Coordinated Plan on AI, reviewed in 2021, represents a joint commitment by the European Commission and Member States to maximise Europe's potential in AI. It focuses on accelerating investments, implementing AI strategies and programs, and aligning AI policy to address global challenges**. The new AI innovation package presented in January 2024 aims to further accelerate investments and assist EU Member States in developing coherent AI strategies**[8]. The EU's initial lead in AI policy was hindered by delays caused by additions to the regulatory text, allowing other global players like China and the US to advance their AI regulations rapidly. These developments shifted the focus from rigid regulations to softer forms of intervention, such as codes of conduct. The co-legislators faced challenges in balancing comprehensive problem-solving with the agility required in fast-evolving AI fields.

Several key initiatives have been launched to support AI development in Europe, as summarised in Figure 4:

- The setup of AI Factories, a new pillar for the EU's supercomputers Joint Undertaking dedicated to acquiring, upgrading and operating AI-dedicated supercomputers to enable fast machine learning and training of large General Purpose AI (GPAI) models and facilitation of the access to the AI dedicated supercomputers, contributing to the widening of the use of AI to a large number of public and private users, including startups and SMEs; and offering a one-stop shop for startups and innovators,

- A decision to establish an AI Office within the Commission, which will ensure the development and coordination of AI policy at the European level, implement the regulation of general-purpose AI systems, supervise the implementation and enforcement of the forthcoming AI Act and foster innovation powered by generative AI.[9]

- The launch of the first two European Digital Infrastructure Consortiums (EDICs), a new instrument made available to Member States under the Digital Decade Policy

[8] European Commission, 2024: Commission launches AI innovation package to support Artificial Intelligence startups and SMEs https://ec.europa.eu/commission/presscorner/detail/en/ip_24_383

[9] European Commission 2024: Commission Decision Establishing the European AI Office: https://digital-strategy.ec.europa.eu/en/library/commission-decision-establishing-european-ai-office

Programme 2030 to speed up and simplify the setup and implementation of multi-country projects by deploying joint infrastructure, delivering services and bringing together public and private entities, end users and industry. The first two EDICs are the Alliance for Language Technologies European Digital Infrastructure Consortium (ALT-EDIC), which seeks to address the scarcity of European language data available for training artificial intelligence (AI) solutions; and CitiVERSE, which connects existing local digital twins across Europe, forming the basis for an EU network.

- Network of Excellence Centres and Public-Private Partnerships (PPPs): These include the ADRA agreement, which plans to raise €2.6 billion for European companies through a mix of public and private investments.

- Testing and Experimentation Facilities (TEFs): With €220 million co-investment, these facilities aim to bridge gaps in the AI value chain and support the deployment of trustworthy AI.

- Horizon Europe (HE) and Digital Europe (DE) Programs, as well as the European Innovation Council Accelerator: These programs focus on advancing large AI models and developing open-source language technologies, with substantial funding allocated for various AI initiatives.

- The European Network of AI Excellence Centres (ELISE), dedicated to connecting high-level research and methods across academia, industry and society[10]

- Public-Private-Partnerships: Agreement to design programs and raise 2.6 billion euros for European companies with the European Association for AI, Data and Robotics (ADRA)[11] with a 1,3 billion euros of public funding to be matched by the private sector for the period 2021-2030

- Testing and Experimentation facilities (TEFs) in co-investment with MS under the Digital Europe Programme (DEP), as world-class reference technology infrastructures at the EU level addressing gaps in the value chain, from the lab, to the market, these would support the deployment and uptake of trustworthy AI and support the implementation of the AI Act, as well as European technology providers of AI solutions

- European Digital Innovation Hubs: networking transfer of expertise: EDIHs will act as a multiplier and widely diffuse the use of all the digital capacities built up under the different specific objectives of DIGITAL. Networking is organised by the Digital Transformation Accelerator, mingling generalists and AI specialists, conceived as a one-stop shop for digital transformation at the regional level. They concentrate services, from basic training to advanced technology deployment, and are open to SMEs and public administrations.

[10] https://www.elise-ai.eu/
[11] https://adr-association.eu/

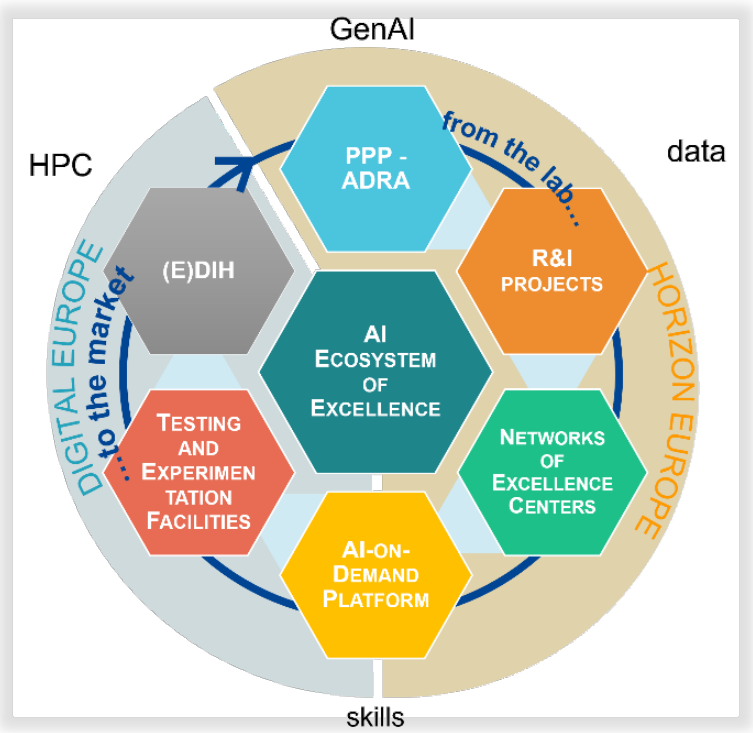


Figure 4 - Articulation of the European Initiatives on AI (Source: European Commission)

2. Perspectives: a European model leveraging European strengths

2.1. EU leadership in science

The EU has a strong position in the world in terms of scientific capabilities, an educated workforce and AI talent enough to develop and harness the power of AI. Figure 5 shows the number of AI-related scientific publications from 2020 to 2024 sourced from the OpenAlex database, the successor to the Microsoft Academic Graph (MAG). It clearly shows the dominant position of China with 20.1% of the AI scientific publications, but in this case, Europe is not lagging behind the US, in fact, it is in front of the US with 18.2% instead of 14.8%. If we include the 6.5% of the UK and EFTA countries, Europe is the world leader in scientific publications in AI.

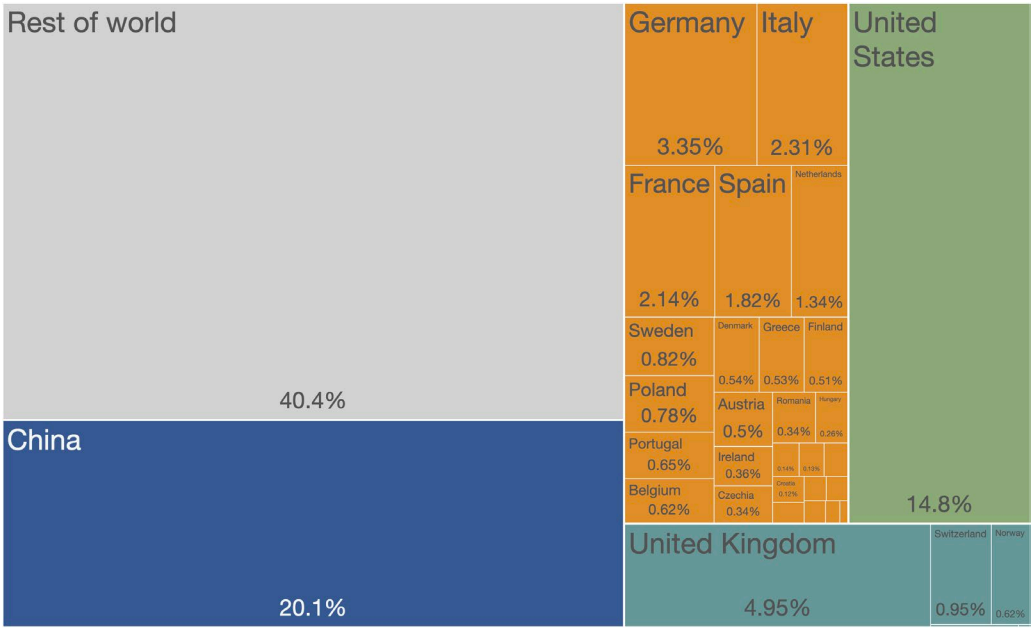


Figure 5. Number of AI scientific publications from 2020-2024 (Source: authors' own work based on OpenAlex)

Europe has a significant opportunity to lead in the application of artificial intelligence (AI) in scientific research, particularly in the field of laboratory robotics. The transformative potential of AI across various scientific disciplines is widely recognized, with AI technology acknowledged as a significant potential game-changer[12]. In this context, the initiatives to leverage European leadership in supercomputing, such as the European High-Performance Computing Joint Undertaking (EuroHPC), are steps in the right direction. These efforts aim to harness the power of supercomputing to advance AI capabilities, thereby reinforcing Europe's position at the forefront of scientific innovation.

Europe is a global leader in AI scientific publications and has solid infrastructures and initiatives, but it also trains a substantial amount of the top AI researchers as shown in figure 6. The Paulson Institute, with its MacroPolo[13] project analyses the trajectory of AI researchers that have presented at the most prestigious AI conference in the world: the 2022 Neural Information Processing Systems Conference (NeurIPS). It shows that the top AI researchers are as likely to have done their undergrad in the combined EU27, UK and EFTA than in the US. What dramatically changes is what happens after undergrad. Almost 14% of the top AI talent that graduated in Europe then went to the US for grad school. This talent is mostly substituted by AI researchers with an undergrad from China (12%). At the grad school level, the US is the place that gains the most, being a magnet from the world's talent (China, India and Europe). After grad school, there is another brain drain from Europe to the US, with more grad talent working in the US than returning to Europe.

[12] European Commission, Directorate-General for Research and Innovation, Petkova, D., Roman, L., *AI in science – Harnessing the power of AI to accelerate discovery and foster innovation – Policy brief*, Publications Office of the European Union, 2023, https://data.europa.eu/doi/10.2777/401605

[13] https://macropolo.org/digital-projects/the-global-ai-talent-tracker/methodology-for-global-ai-talent-tracker-2/

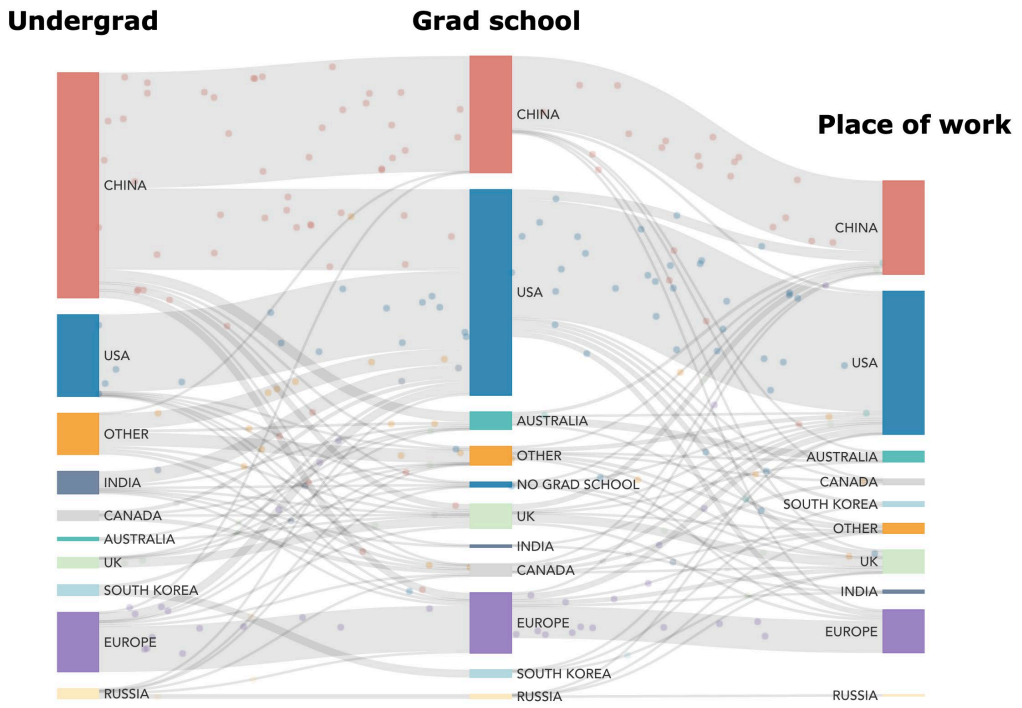


Figure 6. Brain drain and brain gain for top AI researchers (source: MacroPolo, Paulson Institute, edition 2023 based on NeurIPS 2022)

AI is reshaping the skills landscape, emphasizing the growing importance of analytical, social, and domain-specific skills. However, these advancements also highlight significant skills gaps that need to be addressed. The digitalization of education, coupled with the application of AI, offers a transformative potential to enhance the learning experience, making education more accessible, personalized, and relevant to the continuously and rapidly evolving job markets. To redefine the future of education, the European Union will need to take bold actions to integrate digitalization and AI through innovative methods sometimes in tandem with augmented reality (AR), virtual reality (VR), and gamification[14]. These technologies can be instrumental in teaching core skills, including training creativity, by providing immersive, personalized and fully interactive learning environments. This approach not only makes learning more engaging but also ensures that education systems are better aligned with the demands of the modern workforce. By leveraging these advanced technologies, the European Union can foster a more adaptive and forward-thinking educational framework, preparing students for the challenges and opportunities of a disrupted future.

2.2. EU leadership in government technology

The public sector plays a crucial role in the social and economic integration of AI for social good. The emergence of Generative AI (GenAI) models underscores the pressing

[14] European Commission, Directorate-General for Research and Innovation, Dixson-Declève, S., Renda, A., Schwaag Serger, S. et al., *Transformational education in poly-crisis*, Publications Office of the European Union, 2023, https://data.europa.eu/doi/10.2777/98947

need for public sector engagement to harness AI's potential to enhance efficiency and service delivery, particularly in government technology (GovTech). Generative AI models offer transformative possibilities for public services. These models can streamline administrative processes, improve decision-making, and provide more personalized services to citizens. However, realizing these benefits requires active involvement and investment from public authorities to ensure that AI technologies are developed and deployed responsibly and effectively. The EU is actively working in this direction through initiatives such as the Digital Europe Programme. This programme aims to facilitate the uptake of AI by establishing AI hubs that collaborate with the public sector. These hubs serve as innovation centres, aiming at developing and implementing AI solutions tailored to public sector needs. By providing resources, expertise, and support, these hubs help public institutions integrate AI into their operations, thereby enhancing their capabilities and service delivery. The EU's efforts to promote AI adoption in the public sector will provide a more efficient, transparent, and responsive public sector that can better serve its citizens and address societal challenges.

GovTech specifically refers to the use of emerging technologies and digital products and services by government from start-ups and SMEs - instead of relying on large system integrators (such as Google and Microsoft). It entails the following three common elements:

- The public sector engages with start-ups and SMEs to procure innovative technology solutions,
- The procurement enables the provision of tech-based products and services,
- The procurement has at least the goal of innovating and improving public services

The EU stands at the forefront of championing an AI regulatory framework that takes into account people, the planet and prosperity. Such proactive measures could significantly advance human-centric GovTech, placing the needs and rights of citizens above all. This highlights an area where further strategic support and empowerment for smaller business entities are essential but also a new governance model with better public-private goal alignment.

While the importance of global and human-centric regulation of AI is acknowledged, and the EU's specific advances in the sector could be an asset, but also a hindrance to the development of the EU's leadership in that domain. The Single Market has a strong role to play in bringing together the fragmented AI ecosystems and would benefit from a proactive industrial AI policy. State aid (or "European aid") and mission-based policy could contribute to creating an "Airbus moment", following the path of the famous EU industrial leader, while taking into account the constraints of sustainable and socially responsible deployment.

2.3. Self-sovereign infrastructures

Europe has the human, financial and technical resources to be a leader in AI and needs to regain its ability to deploy its own solutions if it doesn't want to fall into the subscription trap. In this context, self-sovereign infrastructure is crucial. Self-sovereign infrastructure refers to systems where users have control over their data and infrastructure, rather than being under government control or dependent on external providers. At the 2023 State of the Union Address, President von der Leyen announced the EU AI Start-Up Initiative, leveraging one of Europe's biggest assets: its public high-performance computing infrastructure. As Europe is a world leader in supercomputing through its European High-Performance Computing Joint Undertaking (EuroHPC), this initiative represents a significant step towards establishing self-sovereign infrastructures. By fostering an environment where

European entities can develop and control their AI solutions, the EU aims to ensure technological independence and enhance its leadership in the global AI landscape.

In order to boost data availability and governance, it is important to take the European approach to data further: to curate and manage larger swaths of data available to AI startups and Small and Medium Enterprises (SMEs). This can occur in many ways: for example, sectoral and horizontal data spaces, still in the making, could be further pushed to enable the aggregation of interoperable data for specific purposes and uses and ensure their availability for SMEs. In addition, public institutions can use the B2G data sharing provisions included in Chapter 5 of the Data Act, and be helped by independent data intermediaries as introduced by the EU Data Governance Act, to then make them available to SMEs (through a GaaP, or "government as a Platform" approach, the GovTech aspect); and the same could be done by institutes of statistics (see Soete 2023). Other recent policy measures that could make more data available to AI developers include the unbundling provisions contained in the Digital Markets Act, and the combination of the digital identity ones in EIDAS 2.0 and the Interoperable Europe Act (still pending).

The European professional AI associations, such as The Confederation of Laboratories for AI Research in Europe (CLAIRE) and the European Laboratory for Learning and Intelligent Systems (ELLIS), have taken steps to unite and support the European Commission in **cultivating a unified European effort in AI and deploy its own solutions**. These efforts need to be met with a coordinated approach integrated with member States' research and industrial policies, including in connecting the interregional AI hubs discussed above (Figure 3). Furthermore, it should ensure that regions where AI innovation cannot take place at a reasonable scale are not left behind from reaping the benefits of European efforts towards AI leadership.

2.4. Rethinking investment

But to become an AI leader, regain competitiveness, and transform research and talent capacity into economic, societal and environmental value there is an urgent need to rethink investment. Europe urgently needs to invest in some of its AI hubs and make sure they grow as funding and talent magnets at a worldwide scale.

The availability of venture capital for AI startups in Europe is a concern raised by AI stakeholders, who highlight the extreme lack of funding and the importance of bolstering public-private investments in AI, as well as the role of public funding in European R&I, especially given the necessity for coherent and coordinated investments, with more focused investments and risk-taking necessary to support the transition, including, for example, local smart specialisation strategies. As of 2020, there was 7 times more venture-capital funding in the US than the EU[15]. And the situation is not improving. Private investment figures in the EU and the US belong to different leagues. In May 2024, Elon Musk's AI company, xAI, has raised $6 billion in a venture capital round that values the new startup at a staggering $18 billion figure.

The few companies that try to build LLMs in Europe, such as Mistral AI or Aleph Alpha, **struggle to raise a similar level of resources, find the right skills, use data in less widely used languages, and gain access to sufficient computing infrastructure**. About 64% of the global AI funding is going to US companies and 16% goes to Chinese companies. Only 5% of the

[15] SRIP 2022 Access to finance : the importance of equity and venture capital https://research-and-innovation.ec.europa.eu/system/files/2022-07/ec_rtd_srip-2022-report-chapter-7.pdf

funding goes to EU27 start-ups. Patent applications over the past 7 years depict a similar picture. 34% of patents come from inventors based in the US, 22% from inventors based in China and only 11% from inventors based in EU27 countries (Figure 7).

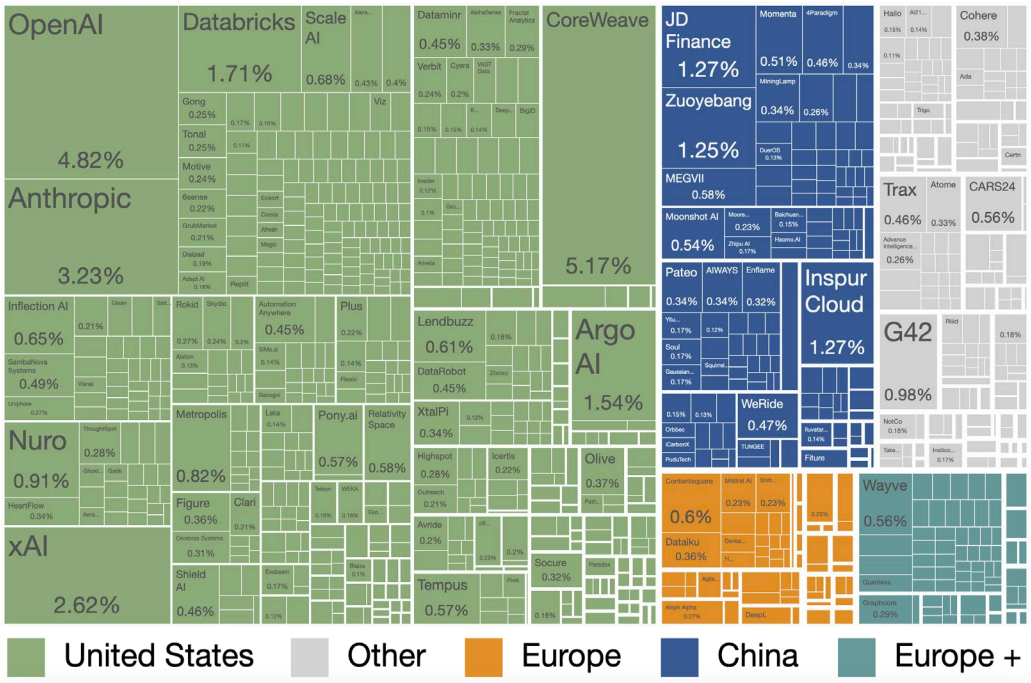


Figure 7 - Share of VC funding received by AI startups (Source: authors' own work based on Crunchbase, version June 2024)

Looking at cities and regions, the picture looks less alarming than at the country scale (see Figure 8). One can see that at the time of writing, the top AI startups in Paris are receiving similar funding to the ones of Boston, and more than in Pittsburgh, Seattle, Chicago, Shenzhen, Guangzhou, or Hangzhou. The gap with the US and China is mainly explained by Silicon Valley, as well as Beijing. But Paris, London, Berlin, Helsinki and Munich also have the potential to lead the AI revolution. A good example is Paris-based Mistral AI, which released Mistral 7B after raising $113 million in June 2023. Not only does this mean serious competition for ChatGPT and Llama3: but it also offers Europe the chance to create a truly open generative AI community. Europe needs to double down on investments and scale them up from regional or European hubs into worldwide success stories.

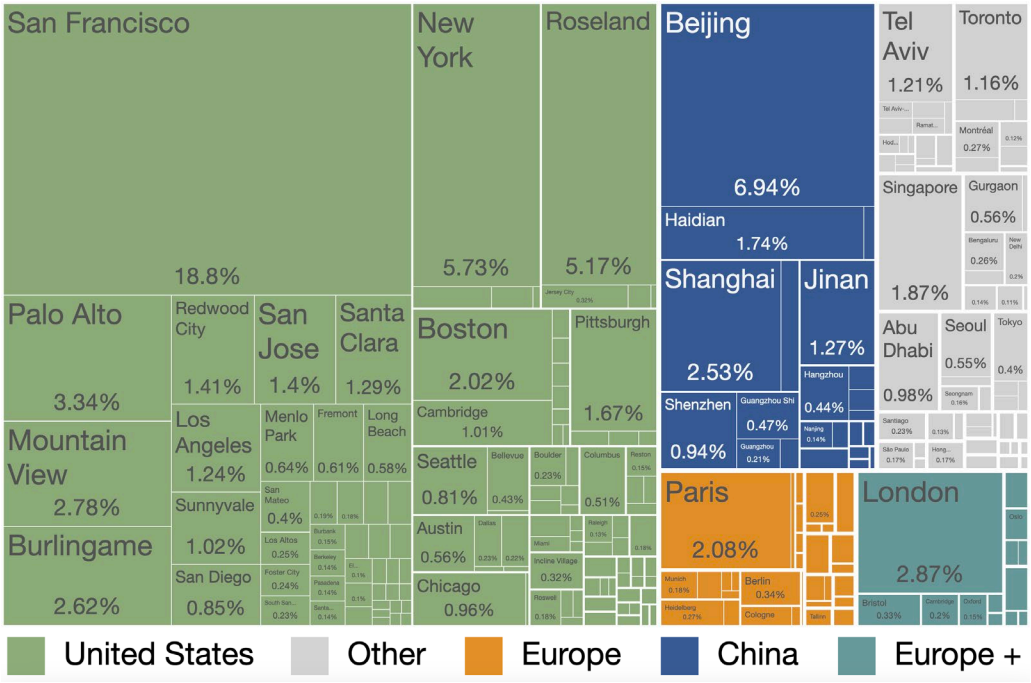


Figure 8 - Share of AI VC funding at the city level (Source: authors' own work based on Crunchbase, version June 2024)

2.5. An innovation-friendly legal framework that places humans at the centre

As this wave of innovation emerges, the need for a comprehensive regulatory framework that ensures the ethical, secure, and equitable use of AI for all should remain a key priority for the EU.

Numerous stakeholders—governments, international organisations, NGOs, and tech giants, are scrambling to address the myriad challenges posed by AI, and each approach has gaps. The European Commission is pioneering efforts to craft the first-ever legal framework for AI. The legislation establishes different rules for different risk levels and has the potential to address AI risks for society. Yet, it is uncertain whether this European effort can address all current and especially future challenges. Two glaring gaps persist in the European legislative effort but also in numerous parallel national or international initiatives.

First, the vast majority of the current efforts are focused on the present and the impacts of narrow AI, that is the current version of AI tools capable of performing specific tasks (like ChatGPT, AlphaFold or AlphaGo). Yet, this preoccupation with narrow AI obscures the potentially catastrophic challenges presented by Artificial General Intelligence (AGI). AGI represents a form of AI with the capacity to comprehend, learn, and apply knowledge across a wide range of tasks and domains. An AGI system connected to the internet and myriad sensors and smart devices could solve complex problems, seek information by any means (even directly interacting with humans), make logical deductions, and even rewrite its own

code. AGI does not exist today, yet according to estimations by foresight experts could be available between 2035-2040, a timeline that coincides with the typical time needed to solidify a global AGI treaty and governance system. This synchronicity underscores the pressing need to pivot European focus, infusing foresight methodologies to discern and tackle imminent challenges and prepare for unknown ones. It is important to note that the latest version of the AI Act goes in the desired direction, as it empowers the AI office to oversee General Purpose AI systems.

The second challenge for the ongoing legislative efforts is fragmentation. AI systems, much like living organisms, transcend political borders. Attempting to regulate AI through national or regional efforts entails a strong potential for failure, given the likely proliferation capabilities of AI. Major corporations and emerging AI startups outside the EU's control will persist in creating new technologies, making it nearly impossible to prevent European residents from accessing these advancements. Additionally, Europe's pursuit of continent-wide regulation poses challenges to remaining competitive in the global AI arena, if the sector enjoys a more relaxed regulatory framework in other parts of the world. Furthermore, Article 6 of the proposed EU Artificial Intelligence Act introduces provisions for 'high-risk' AI systems, requiring developers and deployers themselves to ensure safety and transparency. However, the provision's reliance on self-assessment by AI providers raises concerns about its effectiveness, and requires extensive monitoring and enforcement.

In this light, several stakeholders suggest that any policy and regulatory framework for AI must be established on a global scale. A global technology observatory has been already suggested by the Millennium Project, the Carnegie Council for Ethics in International Affairs and other experts. Such an organisation could be empowered to supervise AI research, evaluate high-risk AI systems, and grant ISO-like certifications to AI systems that comply with standards. It would track technological progress and employ foresight methods to anticipate future challenges, particularly as AGI looms on the horizon. Such an entity, perhaps aptly named the International Science and Technology Organization (ISTO), building on the work done by ISO/IEC and the IEEE on ad hoc standards, could eventually extend its purview beyond AI, encapsulating fields like synthetic biology and cognitive science. Avoiding the usual challenges—dissent over nuances, apprehensions of national sovereignty, and the intricate dance of geopolitics, could be done through the emergence of such an organism from the mentioned already extant standardization organisations. However, the EU, with its rich legacy, is perfectly poised to champion this cause, in close collaboration with the UN to expedite its realisation.

This perspective is supported by numerous ongoing initiatives aimed at fostering international cooperation and standardization in AI governance. For instance, the United Nations has been actively involved in discussions about the ethical and societal implications of AI, advocating for a cohesive global approach. Similarly, the G7 has emphasized the importance of international collaboration in AI development and regulation, recognizing the need for shared principles and guidelines. The Global Partnership on AI ("GPAI")[16] is another significant initiative, bringing together experts from various countries to promote responsible AI development and use. The Bletchley Declaration, an agreement among several nations, highlights the commitment to international cooperation in AI governance[17]. These initiatives recognize that AI, particularly AGI, poses challenges and opportunities that transcend national borders, requiring an urgent and unified global response. Establishing a global

[16] https://gpai.ai/

[17] The Bletchley Declaration by Countries Attending the AI Safety Summit, 1-2 November 2023; https://www.gov.uk/government/publications/ai-safety-summit-2023-the-bletchley-declaration/the-bletchley-declaration-by-countries-attending-the-ai-safety-summit-1-2-november-2023

regulatory framework for AI would ensure that the development and deployment of AI technologies are aligned with shared values and ethical standards, mitigating risks and maximizing benefits for our species.

For the European Commission, initiatives to address AI challenges should not be viewed as mere regulation but as a unique opportunity to etch its commitment to a secure, ethical AI future into history. A long-termist perspective in AI matches with the idea of 'AI Alignment' put forth by numerous scholars, which addresses diverse concerns related to AI safety, aiming to ensure that AI remains aligned with European objectives and avoids unintended consequences of going astray. At the same time, such a long-termist view should not obfuscate the EU's commitment to seizing the AI opportunity and mitigate all challenges posed by the "multi-risk world" (Gürtler 2024).

3. The role of European R&I policy in supporting an AI sector that can deliver on societal goals

A coordinated industrial policy must be put in place to ensure that the comparative advantages of different AI hubs are merged into a consistent picture rather than divided between Member States, supporting systemic industrial transformation in Europe. As discussed by Enrico Letta in his report on the single market and by Fuest, Gros, Mengel, Presidente and Tirole in their middle technology trap report, the fragmentation of the EU system of innovation is a general problem. But probably more than for any other technology, the fragmented nature of AI development across different Member States leads to costly inefficiencies, lack of economies of scale and knowledge spillovers, missed opportunities, and duplications of efforts. Europe needs to build and combine the comparative advantages of various AI hubs to create a more robust and competitive AI ecosystem. This approach would not only enhance EU's technological capabilities but also ensure that the benefits of AI are distributed more evenly across different regions.

As discussed before in this report, one of the key elements of this coordinated policy is the development of EU-wide computing infrastructure. Without adequate computing resources, cutting-edge research ideas cannot become valuable technologies, products or services. In addition to compute infrastructure, a key pillar of this moonshot will have to be the availability of adequate data high-quality and diverse datasets that are essential for training AI models that are accurate and unbiased. Therefore, a coordinated policy should also focus on creating and maintaining data repositories that are accessible to researchers and developers across Europe. This would not only accelerate the pace of AI development but also ensure that the resulting technologies are more reliable and fairer.

But at the end, it all comes down to the talent needed to develop these complex technologies. And as we saw, Europe has a rich pool of talent, but this talent tends to migrate towards the US and is often scattered across different countries and institutions with relatively low mobility. We need bold initiatives, such as the ERC for research and new forms of academic entrepreneurship to retain talent. A coordinated policy should also aim to attract global talent by creating a special immigration policy at the EU level and not at the member states' level (such as the H-1B in the US) and adapting labour laws so that we can create a more integrated talent ecosystem, where foreign researchers, developers, and engineers can easily move, collaborate and share knowledge.

Overall, R&I efforts are often siloed into national AI strategies and lack the critical mass needed to make a global impact. A coordinated policy should aim to create a more cohesive AI policy for the EU, where different institutions and organizations can work together on large-scale, mission-oriented projects. This would not only enhance the quality and impact of European AI research but also ensure that it addresses the most pressing societal challenges. The call for an "Airbus moment" in AI is a call for bold, visionary action. **Just as the creation of Airbus transformed the**

European aerospace industry, a coordinated industrial policy for AI has the potential to transform the European tech landscape. It is an opportunity to harness the collective strengths of different Member States and create a more competitive and innovative Europe. The 2023 State of the Union speech by President Von Der Leyen highlighted the urgency and importance of this task. This strategic vision should also consider the current context of geopolitical competition, security and environmental crises this technology could contribute to. It is now time to turn this vision into reality.

This R&I strategy should be linked to the formulation of sustainable policies and investments, which account not only for the decarbonisation element but also for the need to create "good jobs" (on this see Renda, Balland & Bosoer; 2023). [18] An example of this commitment is the European AI on Demand platform (AIoD) launched by ADRA to bridge European AI research, industry and public services, but Europe can go further.

4. Conclusions

There is a consensus on the urgent need for a cohesive European response to the challenges posed by AI. Whether it is structural issues in the fragmentation of research and deployment of AI or in facing external challenges vis-à-vis our dependence on US and Chinese AI and data systems, Europe needs a unified and systemic response to anticipate the socioeconomic changes that AI will bring to our economy and society.

Europe is well-equipped to tackle these issues, and can still position itself if not as the leader on AI, at least as an autonomous player capable of devising its own solutions and not be dwarfed by the competition. A solid pool of researchers and innovative companies as well as powerful infrastructures in key areas form a core that the EU can build on to strategically leverage AI for the societal good. AI presents an enormous potential in science, government, and industry, but the EU public sector needs to use its unique position to develop and supporting a strategic, transborder and cooperative approach to AI development in Europe, through GovTech and an active role in shaping the new industrial strategy.

Despite this solid core, Europe should enhance its efforts to attract and retain talent, develop the necessary AI infrastructures, reform higher education systems, and support the interconnection of regional AI hubs across borders. Additionally, Europe should further promote the digital transformation of the economy and society, while considering social and environmental constraints.

Data is central to this effort, as its large-scale availability is crucial for the performance of LLMs, yet it also raises privacy and security concerns. The EU must take a proactive role in data utilization, going beyond mere regulation, to ensure that Europeans can actively benefit from their own data.

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[18] Renda, Andrea and Balland, Pierre Alexandre and Bosoer, Lucia, The Technology/Jobs Puzzle: A European Perspective (February 28, 2023). Available at SSRN: https://ssrn.com/abstract=4372626 or http://dx.doi.org/10.2139/ssrn.4372626

Funding and investment are critical issues: there is a strong need for both increased and more focused funding. The additional volume should match the scale of private funding in the US and public investments in China. Failing that, in the case of a budget-neutral strategy with no increase of funding for example, targeted funding is essential to limit the damage. Such targeted funding in specific areas and geographies could help bridge the volume gap between Europe and its competitors, but would not be sufficient to ensure European leadership. This includes venture capital with a risk-seeking approach to innovation and infrastructural investment, which would create favourable conditions and encourage a shift in mindset among private funders, who often follow public initiatives. Targeted investments would also support the connection of regional hubs, as not all European regions are equally equipped to advance AI and its adoption. By specializing in complementary technologies and enhancing their absorption capabilities, the EU can ensure that regions unable to specialize in AI still benefit from the transformation.

To achieve the desired goals, two complementary policy elements can be used: the "moonshot" or "supply-side" approach to AI, and the "demand-side" approach. The supply-side approach, often referred to as the "moonshot" strategy, focuses on developing the technological, infrastructure, and workforce capabilities necessary for AI advancement. This involves massively investing in cutting-edge research and development to push the boundaries of AI EU technology, infrastructure enhancement, and workforce training. The demand-side approach aims to support AI industrial applications and needs based on the societal goals they contribute to. This approach can build on several instruments: leverage public procurement to create a market for AI solutions that address societal challenges, implement regulations that promote a human-centric approach to technology, and provide financial incentives, such as grants, tax breaks, and subsidies, to encourage businesses to adopt AI technologies that contribute to societal goals. The EU needs to integrate both approaches to create a balanced and comprehensive strategy for AI development.

This way, **Europe can lead rather than follow, shaping a future where technology serves humanity rather than the other way around**. Regulation alone is not enough; the EU must accelerate the development of European AI innovation that the world adopts. This approach will align technological advancement with humanitarian principles, establishing a global blueprint for others to follow. It would ensure EU technology sovereignty as well as support the digital transition and the retention of talent, while developing the EU economy and advance policymaking.

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There is a consensus on the urgent need for a cohesive European response to the challenges posed by Artificial Intelligence (AI). Increased investment, policy alignment and skill development, are crucial to leverage the emergence of this ground-breaking technology of AI for societal good. The potential of AI in science, government, and industry, underlines the need for the EU public sector to use its unique position in developing and supporting a strategic, transborder and cooperative approach to AI development in Europe, in a unique European model for AI.

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